Project Interim Report

On

ONLINE VOTING SYSTEM
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

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Submitted By

RAJ SONI

ERP -No -11095

Bachelor of Computer Application with AI and ML in Association with Google

Under the Guidance of

Mr. Harsh Multani

Teacher

SCHOOL OF INFORMATION OF TECH NOLOGY

RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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Abstract

The Online Voting System is a modern application developed to simplify and improve the traditional voting process using digital technology. In conventional paper-based voting systems, the process is time-consuming, requires a large amount of manual effort, and is prone to errors such as incorrect counting and duplication. To overcome these limitations, this system provides a secure, fast, and user-friendly platform where users can cast their votes electronically.

In this system, each user is provided with unique access credentials to ensure that only authorized individuals can participate in the voting process. The system allows users to view the list of candidates along with their details before casting their vote. It also ensures that each user can vote only once, maintaining transparency and fairness in the election process.

One of the key features of this system is automatic vote counting, which eliminates human errors and provides accurate results instantly. The results can be displayed in real-time, making the process efficient and reliable. Additionally, the system maintains data securely, reducing the risk of manipulation or loss of information. This Online Voting System is particularly useful for small-scale elections such as those conducted in schools, colleges, and organizations. It reduces paperwork, saves time, and minimizes the need for physical resources. Overall, the system enhances efficiency, accuracy, security, and convenience, making it a practical solution for modern voting needs.

Introduction

The Online Voting System is a digital application designed to conduct elections in an efficient and user-friendly manner using computers. It serves as an alternative to the traditional paper-based voting system, which is often time-consuming, costly, and prone to human errors. By using this system, the entire voting process becomes faster, more accurate, and easier to manage.

In this system, users are provided with access to view the list of candidates and cast their vote securely. To ensure fairness and transparency, each user is allowed to vote only once. The system maintains proper records of all votes and prevents duplication or unauthorized access.

One of the major advantages of the Online Voting System is its ability to automatically count votes and generate results instantly. This eliminates the need for manual counting and significantly reduces the chances of errors. It also saves time and effort, making the election process more efficient.

Objectives

- To make voting easy and fast
- To allow one user to vote only once
- To count votes automatically
- To show results quickly
- To reduce manual work
- To use OOP concepts

Scope

The Online Voting System is mainly designed for conducting simple and efficient elections in schools, colleges, and small organizations. It provides a digital platform where users can securely cast their votes without using paper-based methods. This system ensures that each user can vote only once, making the process fair and transparent. It also automatically counts votes and displays results quickly, reducing manual effort and chances of errors.

In the future, the scope of this project can be expanded by adding a database to securely store all voting and user data. This will improve data management, security, and reliability of the system. Advanced features such as user authentication, encryption, and online accessibility can also be added to make the system more secure and suitable for larger-scale elections.

System Requirements

Hardware:

- Computer / Laptop
- Minimum 4GB RAM
- Internet connection **Software:**
- Java JDK
- IDE (VS Code / Eclipse)
- Operating System (Windows / Linux)

Methodology

The Online Voting System follows a simple and step-by-step process to ensure smooth and secure voting. First, the user starts the system and logs in using valid credentials. After successful login, the system displays the list of available candidates along with their details.

Next, the user selects their preferred candidate and casts their vote. Before accepting the vote, the system verifies that the user has not already voted, ensuring that each user is allowed to vote only once. This helps maintain fairness and prevents duplicate voting.

Once the vote is submitted, it is securely stored in the system. Finally, the system automatically counts all the votes and generates the result. The result is displayed quickly and accurately, reducing manual effort and minimizing errors in the voting process.

Flowchart

Start

↓

Display Menu

↓

User Login

↓

Show Candidates

↓

User Choice (Vote)

↓

Check (Already Voted?)

↓ Yes → Show Message → End

↓ No

Save Vote

↓

Update Vote Count

↓

Show Result

↓

End

· **ClassName** o **1. Candidate Class** o **Variables:**

o name o votes o **Methods:** o addVote()

o o **2. Voter Class**

o **Variables:** o name o hasVoted o **Methods:** o vote()

o o **3. VotingSystem Class**

o **Variables:**

o candidates[] (array of Candidate)

o **Methods:** o displayCandidates()

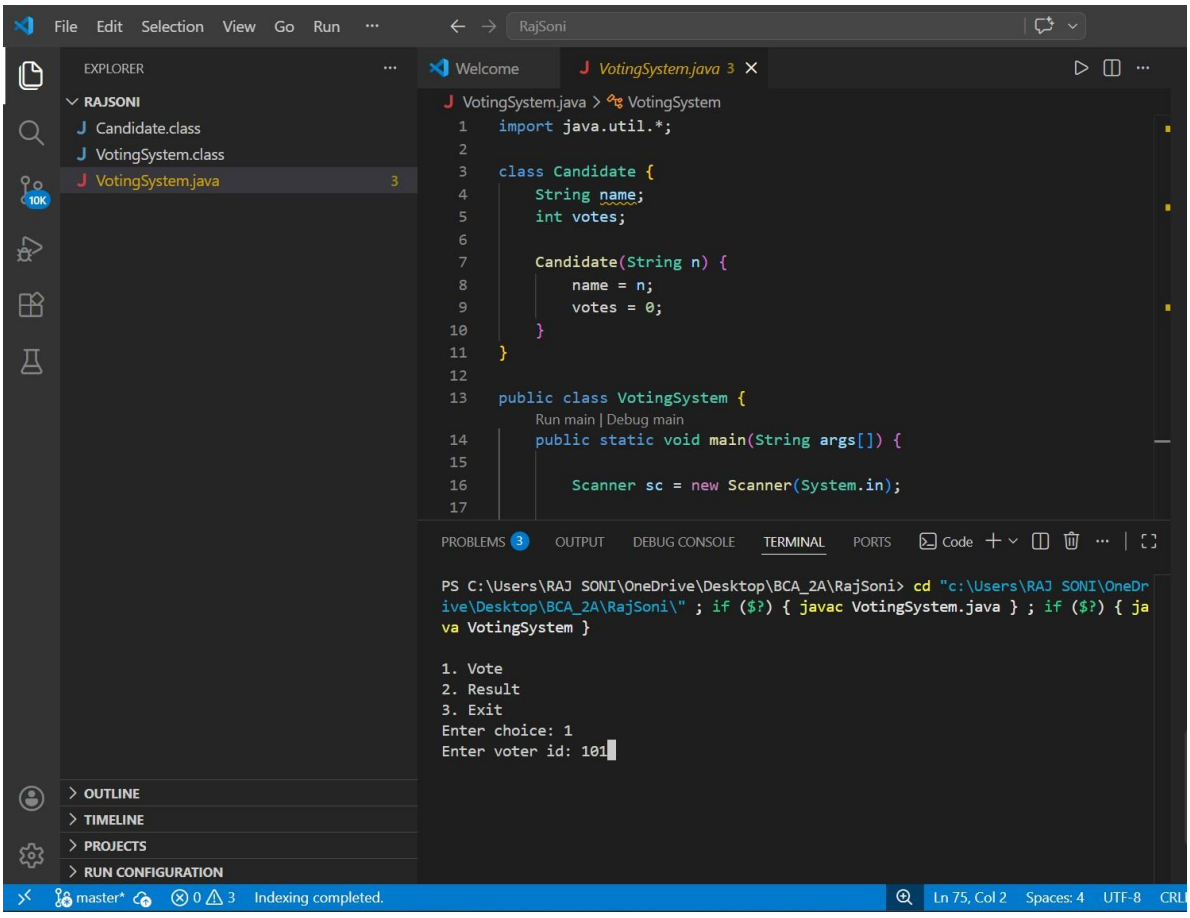
o castVote() o showResult()

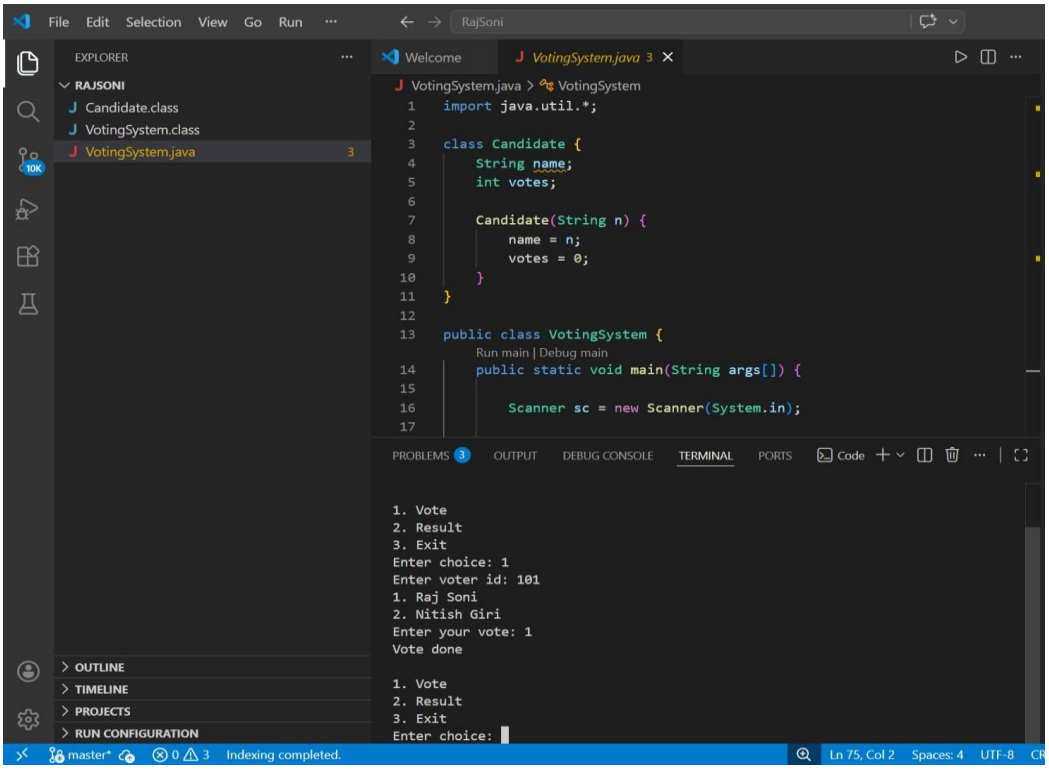
o

Implementation

The project uses different classes. The **Candidate class** stores candidate name and votes. The **Voter class** checks if the user has already voted or not. The **VotingSystem class** manages the whole process like showing candidates, taking vote, saving it, and showing results. It makes sure one user can vote only once.

Sample Input/Output





```java
import java.util.*;

class Candidate {
    String name;
    int votes;

    Candidate(String n) {
        name = n;
        votes = 0;
    }
}

public class VotingSystem {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);
```

```
1. Vote
2. Result
3. Exit
Enter choice: 1
Enter voter id: 102
1. Raj Soni
2. Nitish Giri
Enter your vote: 2
Vote done

1. Vote
2. Result
3. Exit
Enter choice:
```

```
Vote done

1. Vote
2. Result
3. Exit
Enter choice: 2

Raj Soni votes: 1
Nitish Giri votes: 1
Match Draw

1. Vote
2. Result
3. Exit
Enter choice:
```

```
1. Vote
2. Result
3. Exit
Enter choice: 2

Raj Soni votes: 1
Nitish Giri votes: 1
Match Draw

1. Vote
2. Result
3. Exit
Enter choice: 3
Program End
PS C:\Users\RAJ SONI\OneDrive\Desktop\BCA_2A\RajSoni>
```

Future Enhancements

- Add database
- Improve security

- Make it online
- Add GUI

Gantt Chart

A Gantt chart is used to plan and manage the project work step by step. It shows different tasks and the time needed to complete them.

Tasks and Timeline:

- Planning – 1 day
- Designing (Flowchart & Classes) – 2 days
- Coding – 3 days
- Testing – 2 days
- Final Report – 1 day

This helps to complete the project in an organized way and on time.

1. **Time Management**
2. **It shows when each task starts and when it should finish. This helps to complete work on time and avoid delays or last-minute problems.**

Conclusion

The project successfully shows the use of Java OOP concepts. It provides a simple and working voting system. The system makes voting easy, fast, and accurate.

References

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Conference Paper

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Journal Article

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